

This assignment will not be graded and is only for practice.

Key points: 1

Definition of linear dependence: A set of vectors $v_1, v_2, \dots, v_n \in V$ is said to be linearly dependent if there exist scalars $a_1, a_2, \dots, a_n \in \mathbb{R}$, not all zero such that $a_1v_1 + a_2v_2 + \dots + a_nv_n = 0$.

To check the vectors $v_1, v_2, \dots, v_n \in \mathbb{R}^m$ (with usual addition and scalar multiplication) are linearly dependent, we have to verify that the homogeneous system of linear equations $Ax = 0$ has infinitely many solutions, where the j^{th} column of A is v_j .

Definition of linear independence: A set of vectors $v_1, v_2, \dots, v_n \in V$ is said to be linearly independent if $a_1v_1 + a_2v_2 + \dots + a_nv_n = 0$ implies that $a_i = 0$ for $i = 1, 2, \dots, n$.

To check the vectors $v_1, v_2, \dots, v_n \in \mathbb{R}^m$ (with usual addition and scalar multiplication) are linearly independent, we have to verify that the homogeneous system of linear equations $Ax = 0$ has only the trivial solution (i.e., $x = 0$), where the j^{th} column of A is v_j .

If S is a set containing zero vector, then the set S is a linearly dependent set.

Let S be a set of vectors from a vector space V . If an element of S is a scalar multiple of another element from the set S , then S is a linearly dependent set.

1) Vectors $(2,1)$ and $(6, 3)$ from $\mathbb{R}^2$ are

1 point

- ☐ linearly independent.
- ☐ linearly dependent.
- ☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

linearly dependent.

2) Suppose the set $\{(3, 7, 4), v\}$ is a linearly dependent set in the vector space $\mathbb{R}^3$ with usual addition and scalar multiplication. Which of the following vectors are possible candidates for v ? **1 point**

- ☐ $(0, 0, 0)$
- ☐ $(5, \frac{35}{3}, \frac{20}{3})$
- ☐ $(1, 0, 0)$
- ☐ $(6, 14, -8)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(0, 0, 0)$
 $(5, \frac{35}{3}, \frac{20}{3})$

3) The set of vectors $\{(1, 1, -1), (-1, 1, 1), \text{ and } (0, \frac{1}{2}, 0)\}$ in $\mathbb{R}^3$ is

1 point

- ☐ linearly independent.
- ☐ linearly dependent.
- ☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

linearly dependent.

4) $M_{2 \times 2}(\mathbb{R})$ denotes the vector space consisting of real square matrices of order 2 with usual matrix addition and scalar multiplication. Consider the following matrices:

1 point

$$M_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, M_2 = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, M_3 = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$$

Which of the following options is(are) true?

[Hint: Form a system of linear equations using matrix equation $x \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} + y \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = 0$]

- ☐ $\{M_1, M_2\}$ is a linearly independent set.
- ☐ $\{M_1, M_2, M_3\}$ is a linearly dependent set.
- ☐ $\{M_2, M_3\}$ is a linearly dependent set.
- ☐ $\{M_1, M_3\}$ is a linearly independent set.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{M_1, M_2\}$ is a linearly independent set.

$\{M_1, M_3\}$ is a linearly independent set.

Key points: 2

Given two vectors (x_1, y_1) , and $(x_2, y_2) \in \mathbb{R}^2$ construct $A = \begin{bmatrix} x_1 & x_2 \\ y_1 & y_2 \end{bmatrix}$. Then these vectors are linearly dependent if and only if $\det(A) = 0$ and linearly independent if and only if $\det(A) \neq 0$.

Given three vectors (x_1, y_1, z_1) , (x_2, y_2, z_2) and $(x_3, y_3, z_3) \in \mathbb{R}^3$ construct $A = \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix}$. Then these vectors are linearly dependent if and only if $\det(A) = 0$ and linearly independent if and only if $\det(A) \neq 0$

5) Vectors $(2,3)$ and $(4, 5)$ from $\mathbb{R}^2$ are

1 point

[**Hint:** Form the matrix using the vectors and calculate the determinant of the matrix.]

- ☐ linearly independent.
- ☐ linearly dependent.
- ☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

linearly independent.

6) Consider a set of vectors $S = \{(1, 2), (-1, 2), (3, 1), (-3, 1)\} \subset \mathbb{R}^2$. Choose the set of correct options.

1 point

[**Hint:** Form the matrix using the vectors as the columns for each option and calculate the determinant of the matrix.]

- ☐ $\{(1, 2), (-1, 2)\}$ is a linearly dependent set.

- ☐ $\{(-1, 2), (-3, 1)\}$ is a linearly independent set.
- ☐ $\{(-3, 1), (-1, 2)\}$ is a linearly independent set.
- ☐ $\{(3, 1), (-3, 1)\}$ is a linearly dependent set.
- ☐ $\{(3, 1), (1, 2)\}$ is a linearly independent set.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(-1, 2), (-3, 1)\}$ is a linearly independent set.
 $\{(-3, 1), (-1, 2)\}$ is a linearly independent set.
 $\{(3, 1), (1, 2)\}$ is a linearly independent set.

7) The set of vectors $\{(1, 1, 0), (0, 1, 1), (1, 0, 1)\}$ is

1 point

[**Hint:** Form the matrix using the vectors as the columns and calculate the determinant of the matrix.]

- ☐ linearly dependent.
- ☐ linearly independent.
- ☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

linearly independent.

8) Suppose $\{(2, 3, 0), (0, 1, 0), v\}$ is a linearly dependent set in the vector space $\mathbb{R}^3$ with usual addition and scalar multiplication. Which of the following vectors are possible candidates for v ? **1 point**

- ☐ $(2, 3, 1)$
- ☐ $(1, 2, 0)$
- ☐ $(1, 3, 0)$
- ☐ $(0, 1, 1)$

☐ $(\pi, e, 0)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(1, 2, 0)$

$(1, 3, 0)$

$(\pi, e, 0)$

9) Consider three vectors $v_1 = (1, 0, 0)$, $v_2 = (1, 1, 0)$, $v_3 = (1, 1, 1)$ in the vector space $\mathbb{R}^3$, **1 point**
with usual addition and scalar multiplication. Which of the following option is true?

[Hint: Find out the vectors of the given sets in the options and form the matrices using the vectors as the columns and calculate the determinant of the matrices.]

- ☐ $\{v_1, v_2, v_3\}$ is a linearly dependent set.
- ☐ $\{v_1 + v_2, v_1 + v_3, v_3\}$ is a linearly dependent set.
- ☐ $\{v_2 - v_1, v_1, v_2\}$ is a linearly dependent set.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{v_2 - v_1, v_1, v_2\}$ is a linearly dependent set.

Key points: 3

Any set of n vectors in $\mathbb{R}^2$ with $n \geq 3$ is linearly dependent.

Any set of n vectors in $\mathbb{R}^3$ with $n \geq 4$ is linearly dependent.

Any set of $n + 1$ vectors in $\mathbb{R}^n$ is linearly dependent.

Any set containing a linearly dependent set is linearly dependent.

If a set is linearly independent, then every subset of it, is linearly independent.

10) The subset $S = \{(3, 1), (1, 2), (-3, 1)\}$ of $\mathbb{R}^2$ is **1 point**

[Hint: Any set of n vectors in $\mathbb{R}^2$ with $n \geq 3$ is linearly dependent.]

- ☐ linearly dependent.
- ☐ linearly independent.
- ☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

linearly dependent.

11) Set $S = \{(1, 1, -1), (-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2), (1, 0, -2)\}$ subset of $\mathbb{R}^3$ is

1 point

[Hint: Any set of n vectors in $\mathbb{R}^3$ with $n \geq 4$ are linearly dependent.]

- ☐ linearly dependent.
- ☐ linearly independent.
- ☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

linearly dependent.

12) Let S be a linearly dependent subset of $\mathbb{R}^3$. Which of the following options are true?

1 point

- ☐ $S \cup \{(1, 0, 0)\}$ must be linearly dependent.
- ☐ $S \cup \{v\}$ must be linearly dependent for any $v \in \mathbb{R}^3$.
- ☐ $S \setminus \{v\}$ must be linearly dependent for any $v \in S$.
- ☐ There may exist some $v \in S$, such that $S \setminus \{v\}$ is still linearly dependent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$S \cup \{(1, 0, 0)\}$ must be linearly dependent.

$S \cup \{v\}$ must be linearly dependent for any $v \in \mathbb{R}^3$.

There may exist some $v \in S$, such that $S \setminus \{v\}$ is still linearly dependent.

Your score is: 0/12